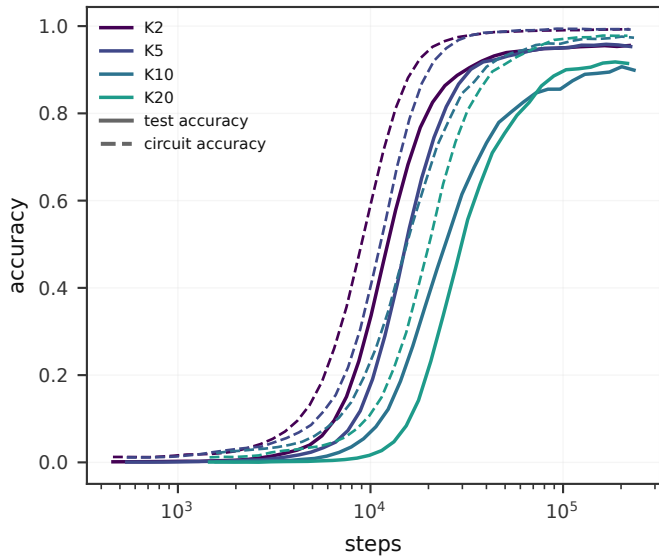
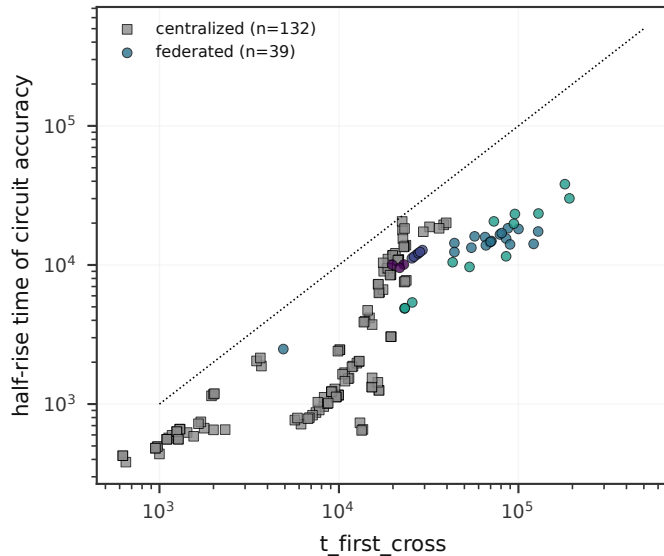


Setup D: the compositional circuit forms long before the full model generalises (grey: centralised)

(a) example federated runs (α 0.3 K ladder, one seed each)



(b) circuit accuracy: median $t_{1/2} / t_{\text{cross}} = 0.33$



(c) irrep structure (U): median $t_{1/2} / t_{\text{cross}} = 0.28$

